10th lesson work – Gravitational Fields

Please see below the assigned work for the 10th lesson. This should take you approximately 1 hour and is designed to be completed in your physics study period – the allocated hour in your timetable with no teacher. You may choose to do this work digitally or on paper; in school or remotely.

12/13

The answers are included, for you to mark your own work. You are expected to show evidence of completion eg workings, as well as submitting your final score out of 13. Please attempt all questions before going ahead to the answers (you will get more out of it if you do!).

1. Kepler-90 is a star with several planets orbiting it.
The two outermost planets are Kepler-90g and Kepler-90h.
Kepler-90g has an orbital period of 210 days and is 0.71 AU from the centre of Kepler-90.
Kepler-90h is 1.01 AU from the centre of Kepler-90.

Kepler's third law of planetary motion can be applied to the planets of Kepler-90.

What is the orbital period of Kepler-90h?

g: 210 0.71 AU
h: 1.01 AU

A 50 days
B 299 days
C 356 days
D 4350 days

$$T^2 = \left(\frac{4\pi^2}{GM}\right) r^3$$

$$\frac{T_g^2}{T_h^2} = \frac{r_g^3}{r_h^3} \Rightarrow T_h = \sqrt{\frac{T_g^2 r_h^3}{r_g^3}}$$

$$= 356.298\ldots$$

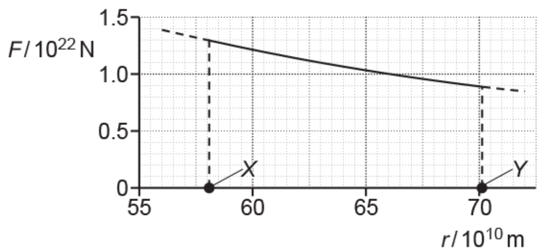
Your answer C

[1]

how is this only 1 mark!!!

2. The planet Mercury has a highly elliptical orbit around the Sun.

The gravitational force F acting on Mercury due to the Sun varies with its distance r from the centre of the Sun. The graph of F against r for Mercury in its orbit is shown below.



$E = F \times distance$

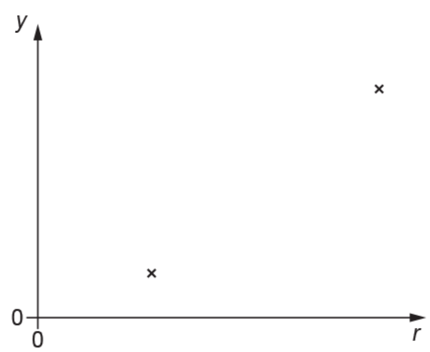
Mercury is closest to the Sun when r = X and furthest when r = Y.

What does the **area** under the graph between the distances X and Y represent?

A The centripetal force acting on Mercury.
B The change in the gravitational potential energy of Mercury.
C The impulse of the force acting on Mercury.
D The kinetic energy of Mercury.

Your answer B **[1]**

3. A student has collected some data on the Solar System.
The student plots a graph, but only two data points are shown below.



The distance from the centre of the Sun is r.

Which quantity y is represented on the vertical axis?

A Speed of a planet. X ↓
B Period of a planet. ✓
C Gravitational potential of the Sun. X ↓
D Gravitational field strength of the Sun. X ↓

Your answer B **[1]**

4. Earth has a mass of 6.0×10^{24} kg and a radius of 6400 km.
A satellite of mass 320 kg is lifted from the Earth's surface to an orbit 1200 km above its surface.

What is the change in the gravitational potential energy of the satellite?

$$-\frac{MmG}{r+1200} - \frac{-MmG}{r} = MmG\left(\frac{-1}{r+1200} - \frac{-1}{r}\right) \times \frac{1}{10^3}$$

$$= 3.159 \times 10^9$$

A 9.1×10^2 J
B 9.9×10^6 J
C 3.2×10^9 J
D 3.8×10^9 J

Your answer C

[1]

5. Algol is a triple-star system, with stars Aa1, Aa2 and Aa3 orbiting each other.
This triple-star is 90 light-years from the Earth.

Here is some data on the star Aa1.

- radius = $(1.90 \pm 0.14) \times 10^9$ m
- mass = $(6.31 \pm 0.42) \times 10^{30}$ kg.

Calculate the gravitational field strength g at the surface of Aa1 to 3 significant figures.
Include the absolute uncertainty in your answer. Assume that the other stars of the system exert negligible gravitational force on Aa1.

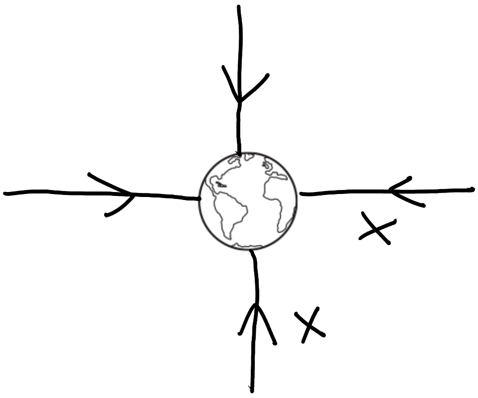
$$g = \frac{MG}{r^2} = \frac{6.31 \times 10^{30} \times 6.67 \times 10^{-11}}{(1.9 \times 10^9)^2} = 116.58\ldots$$

$$\frac{0.14}{1.9} \times 2 + \frac{0.42}{6.31} = 0.2139$$

$$0.2139 \times 117 =$$

g = ...117...... ± ......25.0...... N kg^{-1} **[4]**

6. The diagram below shows the Earth in space.



i. On the diagram above, draw a minimum of **four** gravitational field lines to map out the gravitational field pattern around the Earth.

[1]

ii. On the same diagram above, show **two** different points where the gravitational potential is the same. Label these points **X** and **Y**.

[1]

7. Kepler's third law can be applied to a satellite in a geostationary orbit around the Earth.

i. Complete the equation for Kepler's third law below.
You do not need to define any of the terms.

$$T^2 = \frac{4\pi^2}{GM} r^3$$

[1]

ii. The mass of Earth is 6.0×10^{24} kg.
Calculate the radius of the circular path of a satellite in a geostationary orbit around the Earth.

$T = 60 \times 60 \times 24 \times \cancel{365}$ 86400

$= 31536000$

$r = \sqrt[3]{\frac{T^2 GM}{4\pi^2}} = 6840000$ (3sf)

radius = 6.84×10^6 m **[2]**

geostationary
This is a Satellite (not earth orbitting sun) so it orbits once a day, not year

Mark scheme

Question			Answer/Indicative content	Marks	Guidance
1			C	1	
			Total	**1**	
2			B	1	
			Total	**1**	
3			B	1	**Examiner's Comments** Answering this question needs knowledge of Kepler's Third Law and the formulae for gravitational potential and gravitational field strength, all of which are given in the data, formulae and relationships booklet. This relationship cannot be gravitational potential or gravitational field strength, as quantity y increases with distance from the Sun. By equating the gravitational force on a planet with the centripetal force, it can be shown also that (orbital speed)^2 and orbital radius are inversely proportional. This graph does not show an inversely proportional relationship. The formula sheet says that the square of a planet's period is directly proportional to the cube of the planet's orbital radius. In other words, the relationship shows that as orbital radius increases, so does the period, but not directly proportionally. This is the relationship shown on the graph, giving answer B.
			Total	**1**	
4			**C**	**1**	**Examiner's Comments** This question requires the candidate

					to calculate the gravitational potential energy when r = 6.4 × 10^3 m and again when r = 7.6 × 10^3 m. The difference of those two energies gives answer C. About two thirds of all candidates got this correct.
			Total	**1**	
5			$g = \frac{G \times 6.31 \times 10^{30}}{(1.90 \times 10^9)^2}$ g = 117 (N kg^{-1}) $2 \times \frac{0.14}{1.90} + \frac{0.42}{6.31}$ **or** 0.21 or 21% (absolute uncertainty =) 25 (N kg^{-1})	C1 C1 C1 A1	**Reject** 0.42/(0.14^2) = 21.4 **Note:** final answer of g = 117 ± 25 (N kg^{-1}) with no working scores all 4 marks. **Allow** correct identification of %(r) and %(m) for 1 mark max if no other marks scored. **Allow** alternate method using max/min values of m and r that give the correct absolute uncertainty (28)
			Total	**4**	
6		i	Straight symmetrical radial field lines and correct direction of field	B1	**Ignore** field lines inside the Earth
		ii	X and Y labelled which should be an equal distance away from the centre of the Earth	B1	**Note** Judge by eye **Allow** X and Y both on the surface of the Earth
			Total	**2**	
7		i	$T^2 = \frac{4\pi^2}{GM} r^3$	B1	
		ii	$86400^2 = (4\pi^2/6.0 \times 10^{24} \times 6.67 \times 10^{-11})\ r^3$ (Any subject) radius = 4.23 x 10^7 (m)	C1 A1	**Examiner's Comments** The radius of the orbit of a geostationary satellite was found with ease by the majority of candidates by using the formula in the data book or by looking at their response for part 23(b)(i). There were a few ways of generating an arithmetical error, such as using the square root instead of the cube root for the final step, getting the wrong power for the time or by using a time equal to one year instead of one day.
			Total	**3**	